Saahil Sanganeriya

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Education

Georgia Institute of Technology — *Atlanta, GA*
Master of Science in Computer Science (ML) GPA 4.00 — *Dec 2025*

Georgia Institute of Technology — *Atlanta, GA*
Bachelor of Science in Computer Science (ML, Theory) GPA 3.76 — *Dec 2024*

Experience

Software Engineer, Applied AI — *San Francisco, CA*
Mercor — *Mar 2026 – present*
Team: Data Acquisition, delivering anonymized multi-terabyte enterprise data (billions of tokens) to a frontier AI lab for post-training

- Designed and built entity resolution, which links every name, email and handle to the real person behind it (LLM type classification, FAISS embedding search, Leiden graph clustering), plus its eval suite; raised accuracy on the labeled eval set from **85% to 95%**
- Trained LightGBM and PyTorch match scorers on a 257-feature Rust library, served via ONNX and run in production shadow mode
- Made matching blockwise: MinHash-LSH groups likely matches, one LLM call clusters each block, and merges need agreeing votes (a second mode uses correlation clustering) so one bad verdict can't join two people; lifted B-cubed F1 from **0.34 to 0.84**
- Rebuilt it as a bounded-memory Go service that scales horizontally on Temporal across GKE and EKS; resolved **88M mentions** in one run and halved wall time (**4h13m to 2h11m**) by tuning its LLM judges
- Cut LLM provider failures **22x** (24.6 to 1.1/min, same throughput) over **~3M LLM calls** per run with an adaptive (AIMD) concurrency controller across providers, plus versioned Temporal workers and a custom rollout controller for safe mid-run deploys
- Built the fake-identity generator, giving each person one consistent persona through format-preserving generators for 90+ PII types and length-preserving LLM-written names and addresses; expert reviewers kept **95%** of its replacements on 675 documents
- Created a synthetic-data engine in which an LLM agent designs a fictional company and writes its Slack, Gmail and Linear history while code places every PII value, so labels are exact by construction; its **1.6M labeled mentions** became the team's benchmark
- Redesigned code anonymization to rewrite git histories (tree-sitter, Aho-Corasick) and hand each repo to a sandboxed LLM coding agent that runs its tests; it runs **~200 repos** at once on ECS and cut a 55k-commit repo from **3+ hours to 19 min**

Data Engineer Intern — *Seattle, WA*
Amazon — *May 2025 – Aug 2025*

- Engineered a real-time streaming pipeline (Apache Flink in Java, Kinesis, Lambda) that replaced an hourly batch job for Amazon's workforce management, processing cross-region SNS events at **600–900ms** latency against a 120s SLA
- Compared Firehose, OpenSearch Ingestion, Data Prepper and direct-Lambda designs against Flink on latency, exactly-once guarantees and cost, then rebuilt a PyFlink notebook prototype as a Java DataStream app for multi-sink writes and IAM auth
- Built 5 custom OpenSearch sinks with AWS IAM SigV4 auth and Painless scripts that reject out-of-order events, for exactly-once results with automatic failover and Lambda headroom for 300K+ events/min
- Designed a fault-tolerant dual-path architecture (real-time + S3 backup) spanning AWS regions (Australia to US) with dead-letter-queue replay, retries, CloudWatch alerting and live OpenSearch dashboards; later adopted by other teams for real-time use cases

Data Engineer Co-op — *Atlanta, GA*
The Coca-Cola Company — *May 2023 – Aug 2024*

- Built an enterprise AI chatbot in Haystack with hybrid retrieval (BM25 over Elasticsearch plus a RoBERTa reader), letting Coca-Cola employees search internal knowledge bases and get answers in natural language
- Architected scalable Django REST APIs with CRUD endpoints serving 300+ global supply-chain datasets, replacing manual data requests so BI teams could self-serve operational queries
- Rebuilt a Python CRUD library and monitoring platform, cutting support tickets by **63%** and ETL runtime from **9 to 6.5 hours**
- Built Azure Data Factory ETL pipelines processing 300+ datasets daily via PySpark and SQL on Databricks, cutting transformation-step runtime **60%**, and added CI/CD with automated data validation

Research and Publications

Graduate Researcher — *Atlanta, GA*
Georgia Institute of Technology — *Aug 2024 – Dec 2025*
VandeLoo AD*, Malta NJ*, **Sanganeriya S**, Aponte E, van Zyl C, Xu D, Forest C. *SAMCell: Generalized label-free biological cell segmentation with segment anything.* PLOS ONE, 2025

- Developed SAMCell: proposed its distance-map regression with watershed post-processing and trained it, beating Cellpose and StarDist on LIVECell (**0.77 vs 0.68** OP-CSB) and nearly doubling Cellpose's zero-shot score on an unseen microscope (**0.60 vs 0.32**)
- Fine-tuned SAM's ViT image encoder and mask decoder with a frozen prompt encoder, removing manual prompting for hundreds of cells per image; trained on 5,000+ LIVECell images in PyTorch with CLAHE preprocessing and 256x256 sliding-window inference
- Created two cross-microscope benchmark datasets (PBL-HEK, PBL-N2a) for zero-shot evaluation and shipped an open-source Napari plugin and GUI so biologists without ML expertise can run it

Projects

Deepfake Detection System — *Jan 2025 – May 2025*

- Fine-tuned a GAN-trained ResNet-50 detector on social-media transformations (WebP compression, color jitter, sharpening), lifting accuracy from **51% to 83%** on 19,000 modern deepfakes while keeping fake recall above 93%

Reinforcement Learning Skill Discovery for Atari Games — *Jan 2025 – May 2025*

- Adapted METRA skill discovery to discrete-action Atari with SAC-Discrete, learning distinct skills without hand-designed rewards
- Diagnosed representation collapse with t-SNE and silhouette analysis, and fixed it with a stronger ResNet-18 pixel encoder

Skills

Languages: Python, Go, Rust, Java, SQL, C/C++, JavaScript, Scala
ML & LLMs: PyTorch, LightGBM, ONNX, Hugging Face Transformers, FAISS, scikit-learn, LLM APIs (Claude, Gemini, Bedrock)
Distributed Systems: Temporal, Kubernetes (EKS, GKE), Helm, KEDA, Karpenter, Terraform, Docker, Modal, Spark, Flink, Kinesis
Cloud & Data: AWS (ECS, S3, Lambda), Azure (ADF, ADLS), Databricks, Parquet, DuckDB, PostgreSQL, OpenSearch, Snowflake
Tools: Git, Linux, Datadog, CloudWatch, Airflow, Pandas, NumPy, CI/CD